

Continuous detector systematics for DUNE

A SnowStorm production and analysis chain on the Far Detector, horizontal drift — built, run and analysed end to end.

`[your name]` · `[institute]` · `[date]`

THROWN PER FILE

α recombination ModBoxA

β recombination ModBoxB

τ electron lifetime

991

universes delivered

99,100

atmospheric ν events

justIN workflow 20447 · dunesw v10_16_00d00
seed atmnu-hd-2026a · dial set e1d55d70

Detector systematics, the expensive way

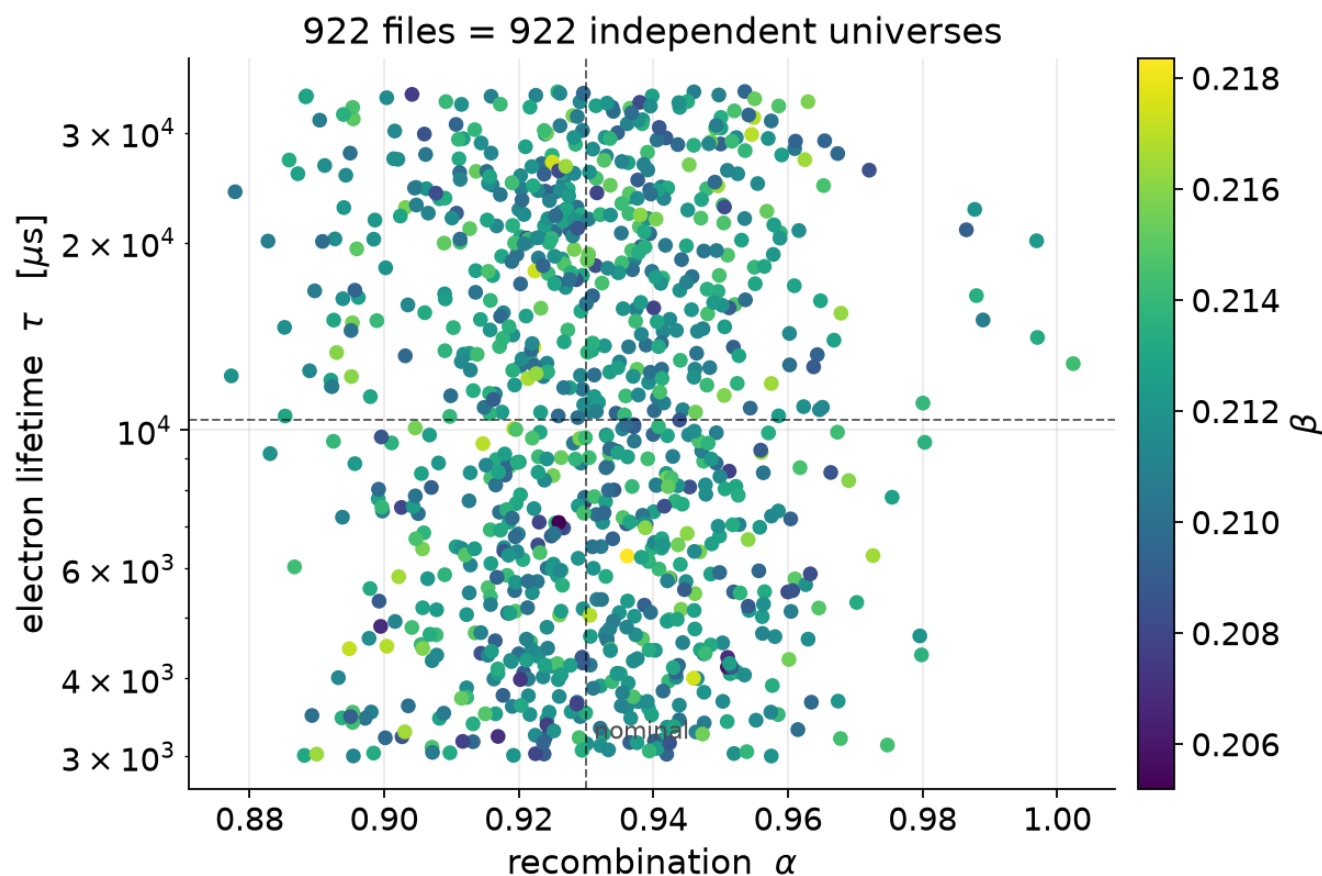
One nominal sample, plus a dedicated sample per dial shifted by $\pm 1\sigma$.

- a **full simulation pass per shifted sample** — cost scales with the number of dials
- only the dials chosen in advance, only at the points chosen
- nothing **between** the points, nothing **off-axis** in the joint space
- correlations between dials are **asserted, not measured**

IceCube solved this with SnowStorm. DUNE's DetSuM effort attacks it with repeated **discrete** points. What follows is the **continuous** complement, built on justIN.

We want the detector response as a continuous function of the parameters — not its value at three points.

One sample, continuously thrown

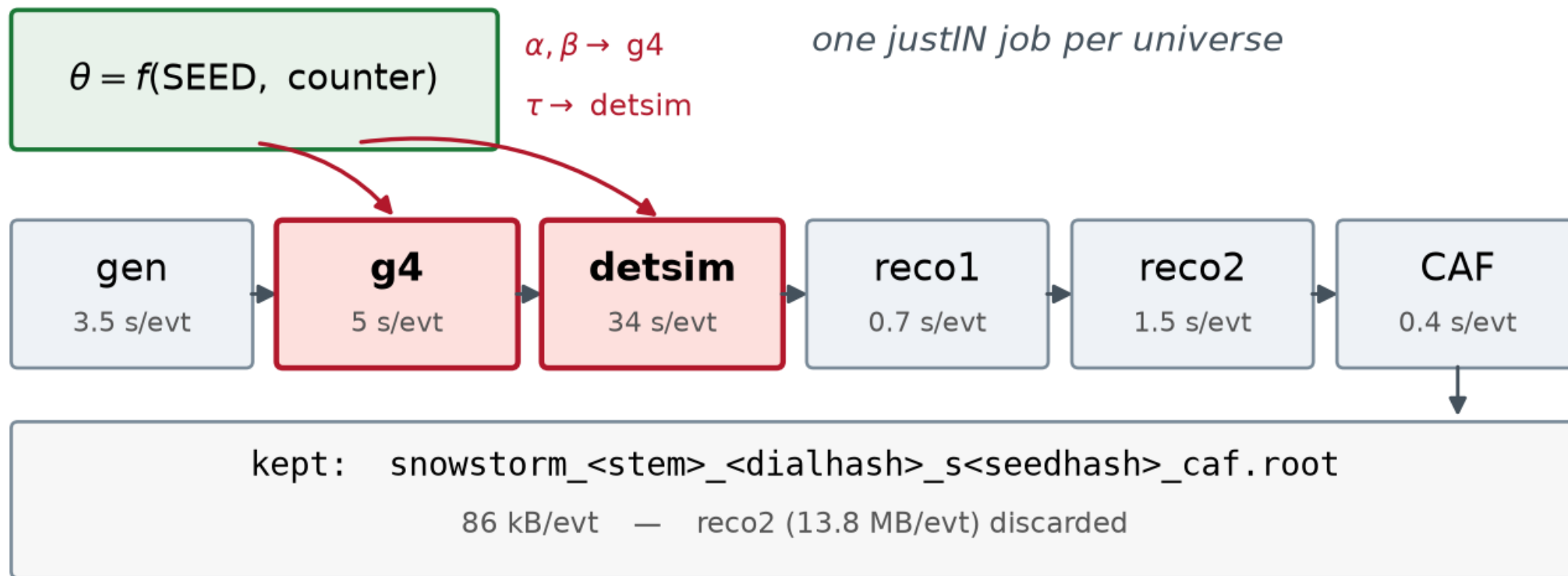


Draw the detector parameters **per file** from a prior, then simulate normally. The ensemble samples the joint distribution $p(E_{\text{reco}}, \theta) = p(E_{\text{reco}}|\theta)p(\theta)$.

The inversion that makes it work: a per-file summary statistic plotted against that file's thrown parameter **is** the response curve. No shifted samples, no reweighting.

922 files $\rightarrow$ 922 independent points in (α, β, τ) . Dashed lines mark nominal.

Every file is its own universe.

**Self-describing**

parameters recoverable **from the filename alone** (stem + dial hash + seed hash); also written to MetaCat as

```
snowstorm.*
```

One job per universe

gen→CAF in a single justIN job; only the CAFs are kept

Bolting it onto a chain you already run

The sampler emits **fcl override lines** — so integration is three lines, not a framework.

YOUR JOBSRIPT

```
# 1. one call defines this job's universe
SSP="python3 snowstorm_params.py \
  --seed $SEED --stem $stem --dials $DIALS"

# 2. prepend the overrides to your own fcl
{ echo '#include "standard_g4_dune10kt_1x2x6.fcl"'
  $SSP --stage g4
} > local_g4.fcl

# 3. run lar exactly as before
lar -c local_g4.fcl -o out.root in.root
```

ADDING A DIAL IS A JSON BLOCK

```
"elecgain": {
  "fcl": "...wcls_main.structs.elecGain",
  "stage": "detsim", "dist": "gaus",
  "nominal": 14.0, "sigma": 0.3,
  "clip": [12.5, 15.5] }
```

RUN IT — JUSTIN PULLS THE JOBSRIPT FROM GITHUB, NOTHING TO CLONE

```
justin simple-workflow --monte-carlo 5 \
  --jobscript-git pgranger23/snowstorm-justin/ \
  examples/minimal.jobscript:v0.2.0 \
  --env SEED=my-first-snowstorm
```

Nothing below the dial file is DUNE-specific: any fcl parameter, at any art stage, can be a dial.

• github.com/pgranger23/snowstorm-justin

What it cost, what it delivered

2 300

core-hours for 1000 × 100 events

5.7 GB

of CAFs — reco2 would be ~940 GB

991/1000

files, all distinct θ

0.3 %

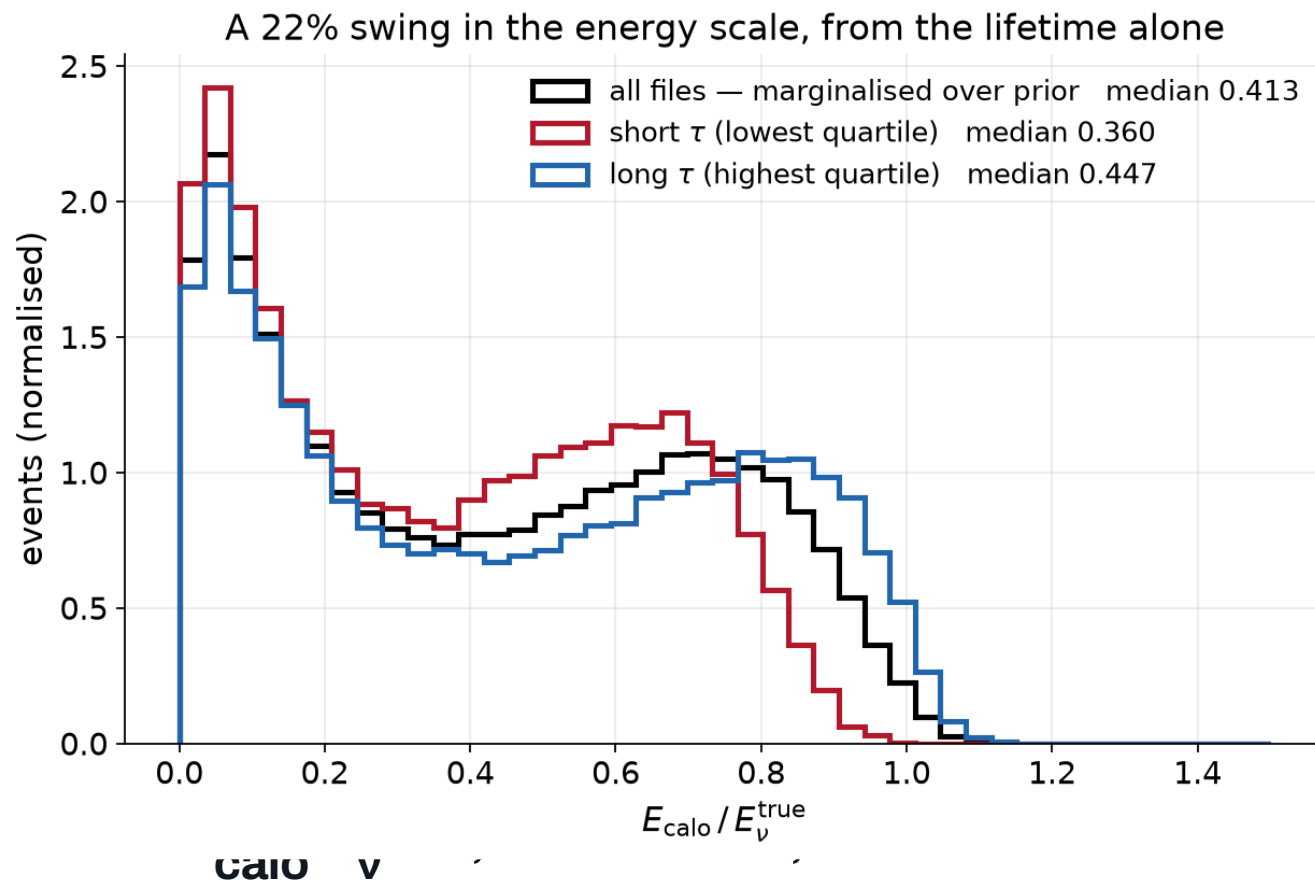
hard job failures

- cost model **measured**: $T(n) \approx 245\text{ s} + 45\text{ s} \times n$ locally, ~80 s/event on the grid (6× site spread)
- **detsim** is 75% of the marginal cost
- slow tail = rare multi-hour shower events → 10 h wall budget
- with `randompolicy` a retry draws **new events at the same θ** , so retries converge

DIAL	PRIOR THROWN	DELIVERED
α	Gaussian 0.93 ± 0.02	0.9303 ± 0.0191
β	Gaussian 0.212 ± 0.002	0.2120 ± 0.0020
τ	log-uniform 3–35 ms	$13.3 \pm 9.1\text{ ms}$, spans 3.0–34.9

Caveat, stated up front: 922 of the 991 are analysed here — 69 sit on replicas unreachable from the gpvm (CCIN2P3 TLS, NIKHEF token). Their throws are indistinguishable from the analysed set (-1.1σ , -1.1σ , -0.3σ in α , β , τ): the loss is **unbiased, not a coverage hole**.

The systematic is visible in the raw spectrum



Splitting the **same ensemble** by thrown lifetime moves the reconstructed energy scale by 22%.

short τ 0.360

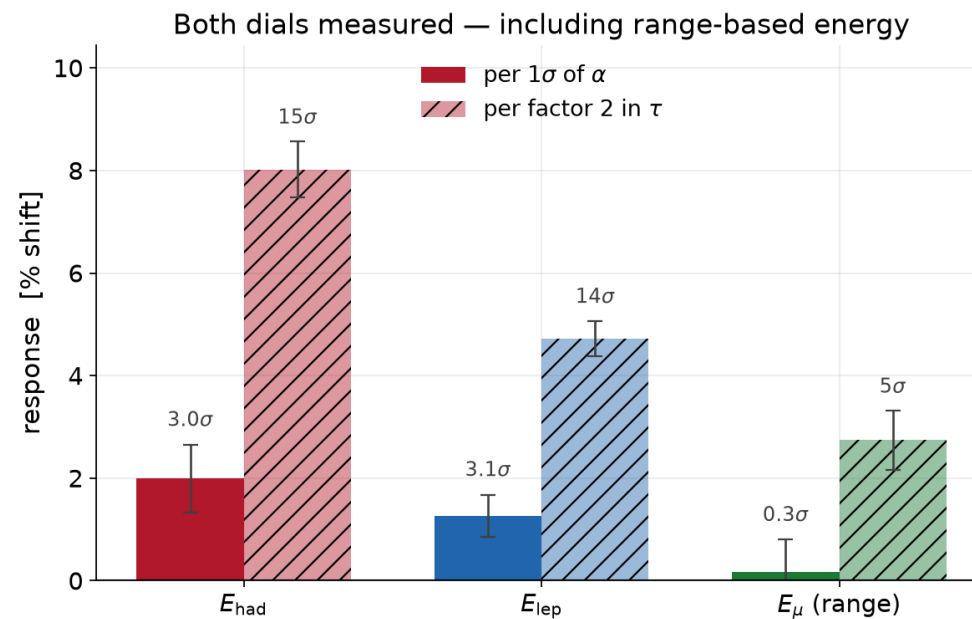
all 0.413

long τ 0.447

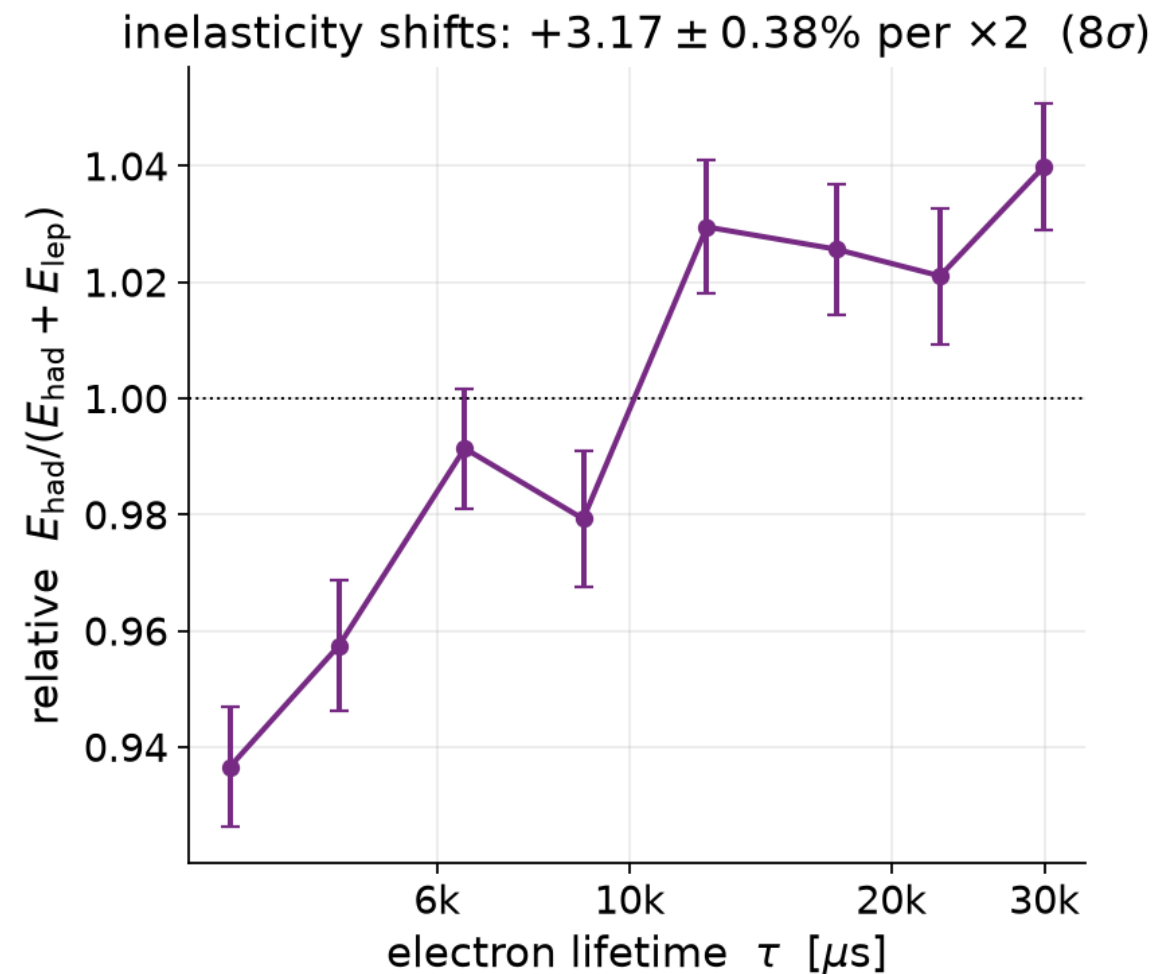
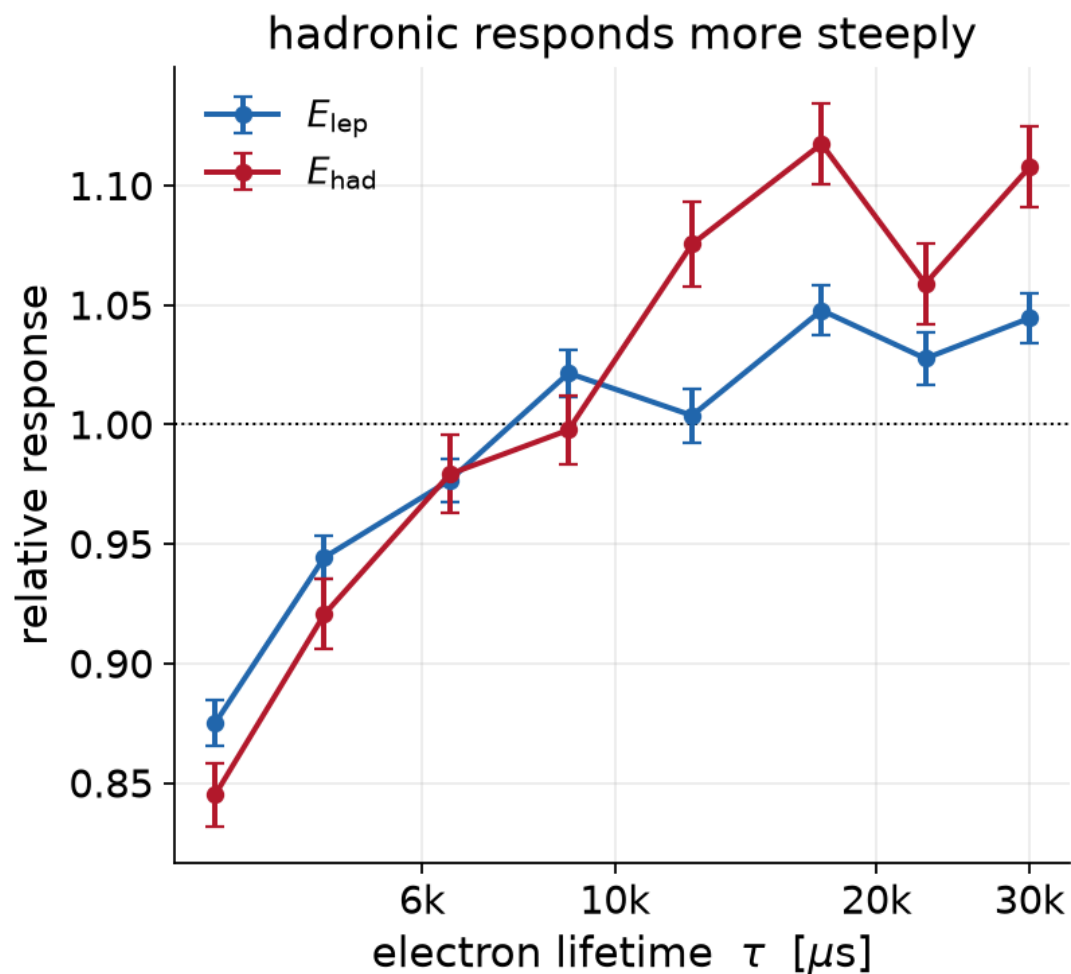
No special analysis — just a cut on a number carried in the filename.

Both dials measured — and one surprise

OBSERVABLE	PER 1σ OF α	PER FACTOR 2 IN τ
E_{had}	$+1.99 \pm 0.66 \%$ (3.0σ)	$+8.02 \pm 0.55 \%$ (14.7σ)
E_{lep}	$+1.26 \pm 0.41 \%$ (3.1σ)	$+4.72 \pm 0.35 \%$ (13.6σ)
E_{μ} from range	$+1.29 \pm 0.45 \%$ (2.9σ)	$+4.64 \pm 0.38 \%$ (12.2σ)

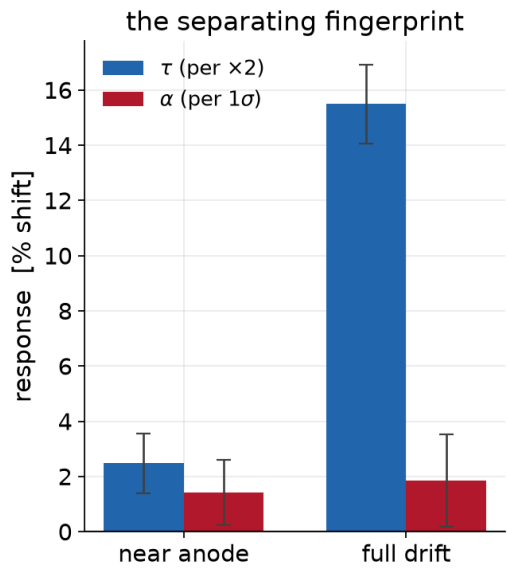
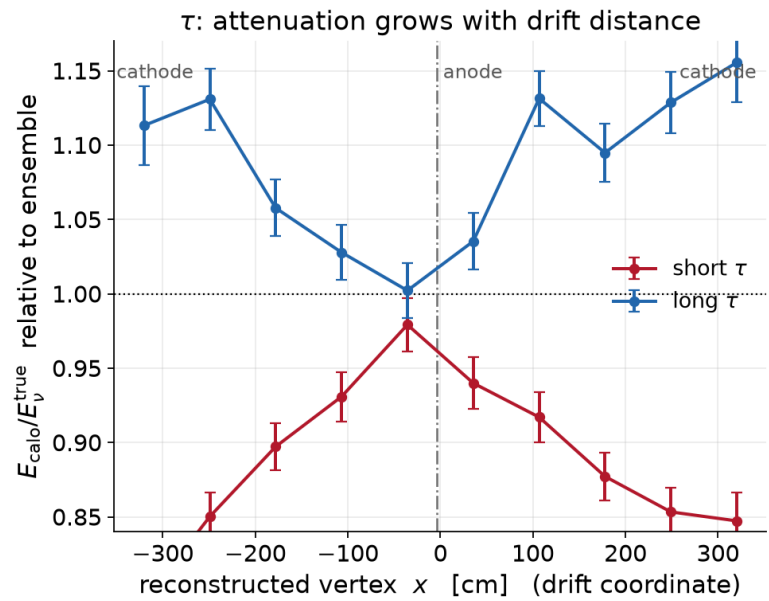


Range-based muon energy is not immune. Range is geometric, so this is not a charge-scale effect: attenuated charge at long drift falls below hit threshold and tracks reconstruct **short**. Not a selection effect either — efficiency moves only 0.7% per factor-2 τ ($91.1\% \rightarrow 93.0\%$).



A pure energy-scale error would cancel in this ratio. It does not — which matters for any analysis binning in inelasticity, or using it for $\nu/\bar{\nu}$ separation.

The fingerprint that separates the dials



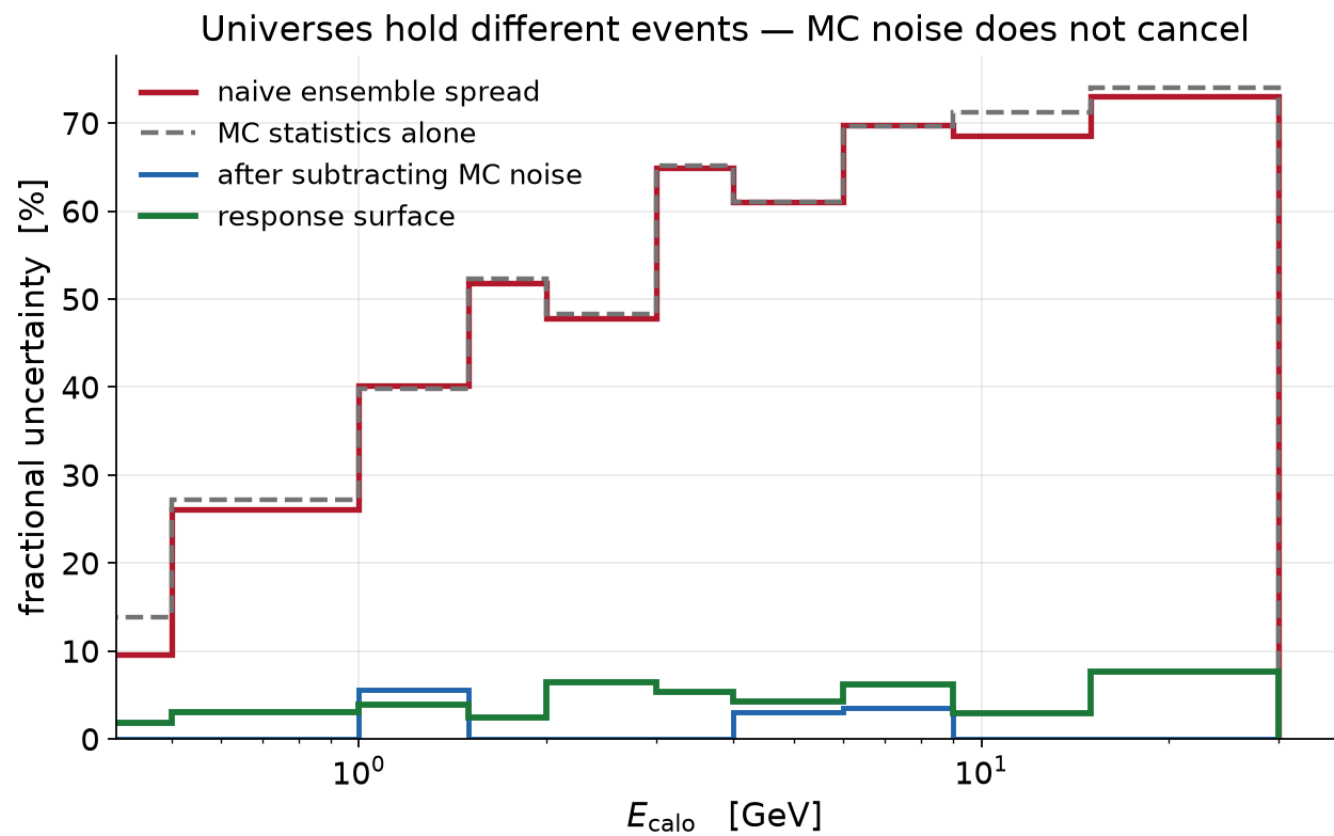
Lifetime attenuates as $e^{-t/\tau}$, so its effect **grows with drift distance** — a clean V centred on the anode, symmetric to both cathodes.
Recombination is a uniform charge-scale error: **flat**.

	NEAR ANODE	FULL DRIFT
τ (per $\times 2$)	$+2.48 \pm 1.09 \%$	$+15.49 \pm 1.44 \%$
α (per 1σ)	$+1.42 \pm 1.19 \%$	$+1.85 \pm 1.66 \%$

The whole-detector number (+8%) **understates** the effect at full drift (~+16%).

The two dials are not degenerate — a fit can separate them from where the event happened.

Turning the ensemble into an analysis input



The trap: in an ordinary multisim, universes are *reweightings of the same events*, so MC noise is common and cancels. Here every file holds **different events** — it does not cancel.

E_{CALO}	NAIVE	MC STAT	SUBTRACTED	RESP. SURFACE
0.5–1 GeV	26.1	27.1	0	3.0
2–3 GeV	47.8	48.4	0	6.5
15–30 GeV	73.0	74.0	0	7.6

At ~92 events/universe the naive covariance overstates the systematic ~20×; subtracting leaves a difference of two large noisy numbers. Regressing across all 922 universes recovers it cleanly.

Closure: calibrate on half the sample, recover the throw on the other half — $\chi^2/\text{dof} = 2.4$, no bias.

What we learned, what to run next

The method works and the chain is production-ready.

- **Throw α wider — the main recommendation.** The $\pm 2.2\%$ Gaussian prior needed ~ 900 universes to reach 3σ . A **wide uniform** throw, reweighted afterwards to whatever prior the systematics group settles on, makes the sample reusable and the response measurable. One line in `dials.json`.
- **More files beats more events per file** — same CPU, better coverage.
- **Counter-derived generator seeds** unlock paired sampling: the same events at two θ , MC noise cancelling in the derivative. `randompolicy` forecloses this today; changing it costs nothing.
- **Next dials:** DL/DT, electronics gain — and the same chain on VD.
- **Priors need signing off** by the systematics group: the current ones are reverse-engineered from DetSuM grid spacing.

• github.com/pgranger23/snowstorm-justin

code, dial registry, jobscrips, the tutorial and every figure in this talk · [docs/ATMNU_HD_PRODUCTION.md](#)

Measured per-stage cost

dunesw v10_16_00d00, geometry v6, one core, HD 1×2×6 atmospheric events.

STAGE	FCL	S / EVENT	OUTPUT / EVENT
gen	prodgenie_atmnu_max_weighted_randompolicy_dune10kt_1x2x6.fcl	3.6	12 kB
g4	standard_g4_dune10kt_1x2x6.fcl	5.0	6.2 MB
detsim	standard_detsim_dune10kt_1x2x6.fcl	34	11.6 MB
reco1	standard_reco1_dune10kt_1x2x6.fcl	0.7	11.8 MB
reco2	standard_reco2_atmos_dune10kt_1x2x6.fcl	1.5	13.8 MB
CAF	standard_cafmaker + IsAtmoCVN	0.4	86 kB

Grid reality differs from the bench: the canary saw 1313 / 1819 / 2435 s for 20 events at three UK sites, while a production job did 100 events in 2064 s on an AMD EPYC at Sheffield — an **18 → 110 s/event spread across sites**, which is what the wall-time headroom is really for.

Three fcl traps, found by running them

TRAP	SYMPTOM	FIX
_geov5	job aborts: <i>“Geometry used for run ... is incompatible”</i>	drop the suffix — the standard chain is v6 in v10_16_00d00
standard_cafmaker	segfaults on event 1 of atmospheric reco2	<code>physics.analyzers.cafmaker.IsAtmoCVN: true</code>
runreco-... variant	redundant	reco2 already runs ereco / anglereco / CVN — CAF is pure packaging

Operational notes

- reading these CAFs over the `/pnfs` NFS mount hung a reader in **uninterruptible sleep for 35 h** (`timeout` cannot kill D-state) — read through the dCache **xrootd** door instead
- the default DUNE token carries only `storage.read:/resilient/jobsub_stage` ; remote replicas need `htgettoken --scopes="storage.read:/..."`
- `justin show-replicas --mql "..."` gives replica URLs with no X509 proxy

Reproducibility and bookkeeping

Every output file names its own parameter point:

```
snowstorm_dune10kt_1x2x6_000252_e1d55d70_s2ea20f_caf.root
      |           |           |           |
      |           |           |           └─ seed hash
      |           |           └─ dial-set hash (priors + fcl paths)
      └─ stem = sample + counter
```

- the same values are attached in MetaCat as `snowstorm.alpha / beta / etau / key / seed / dial_hash`
- **verified:** every canary file and a random sample of production files match the values predicted **before submission**, to the last digit
- a **changed prior changes the dial hash**, so samples thrown from different priors can never be confused
- `OFFSET` separates canary from production counters — identical counters would otherwise collide in both filename *and* subrun number